

Ex. No. 3**MATRIX OPERATIONS****# Input matrix dimensions**

```
r <- as.integer(readline("Enter number of rows: "))
c <- as.integer(readline("Enter number of columns: "))
```

Create first matrix

```
mat1 <- matrix(0, nrow = r, ncol = c)
cat("\nEnter elements for Matrix 1:\n")
```

```
for(i in 1:r){
  for(j in 1:c){
    mat1[i, j] <- as.numeric(readline(
      paste("Matrix1 [", i, ",", j, "]: ", sep = ""))
    ))
  }
}
```

Create second matrix

```
mat2 <- matrix(0, nrow = r, ncol = c)
cat("\nEnter elements for Matrix 2:\n")
```

```
for(i in 1:r){
  for(j in 1:c){
    mat2[i, j] <- as.numeric(readline(
      paste("Matrix2 [", i, ",", j, "]: ", sep = ""))
    ))
  }
}
```

```
}
```

```
cat("\nMatrix 1:\n")
```

```
print(mat1)
```

```
cat("\nMatrix 2:\n")
```

```
print(mat2)
```

Matrix Addition

```
cat("\n===== MATRIX ADDITION =====\n")
```

```
add_result <- mat1 + mat2
```

```
print(add_result)
```

Matrix Multiplication (only if square)

```
cat("\n===== MATRIX MULTIPLICATION =====\n")
```

```
if(r == c){
```

```
  mul_result <- mat1 %*% mat2
```

```
  print(mul_result)
```

```
} else {
```

```
  cat("Matrix multiplication not possible (rows not equal to columns)\n")
```

```
}
```

Transpose

```
cat("\n===== TRANSPOSE =====\n")
```

```
cat("Transpose of Matrix 1:\n")
```

```
print(t(mat1))
```

```
cat("Transpose of Matrix 2:\n")
```

```
print(t(mat2))
```